DEVELOPMENT OF A SYSTEM FOR FAST VIDEO DECODING IN AV1 FORMAT

Project proposal

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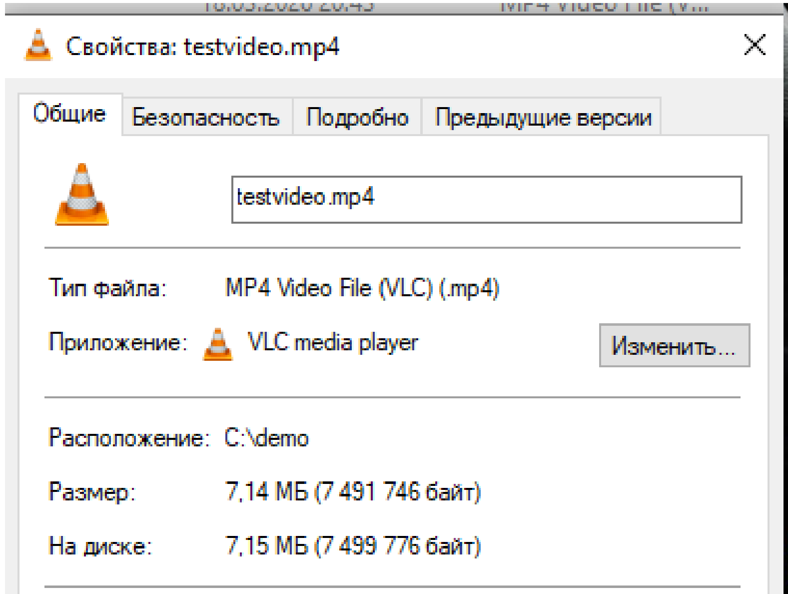
Introduction

- Comparison of an uncompressed video vs compressed: the difference is >300 times!
- The flip side of the coin is high computational complexity
- Although, video decoding is still very fast



Introduction: Theory of Active Perception

- TAPe indexer is something like a one-way video encoder
- Cause: TAPe average time to index is about 24 times faster than to decode a video
- Effect: a large bottleneck in the system

Problem statement

The goal of the project is to develop a system for fast video decoding in AV1 format in 64px resolution.

Delimitations of the study

- The widest spectre of technologies in scope of speed and scalability
  - Frontend
  - Backend
  - **Video decoding (libaom)**
  - Low-level programming
- CUDA disabled
- Single-core
- Ultralow resolution in 64px

Research progress

- ✓Functional and non-functional requirements
- ✓Analysis of the AV1 Decoding specification
- ✓Identification of bottlenecks in the standard
- ✓Identification of all the methods to speed up the code
- ✓Choosing the technology stack
- ✓Designing the naive realization and the optimized realization
- Implementing the realizations
- Testing the realizations

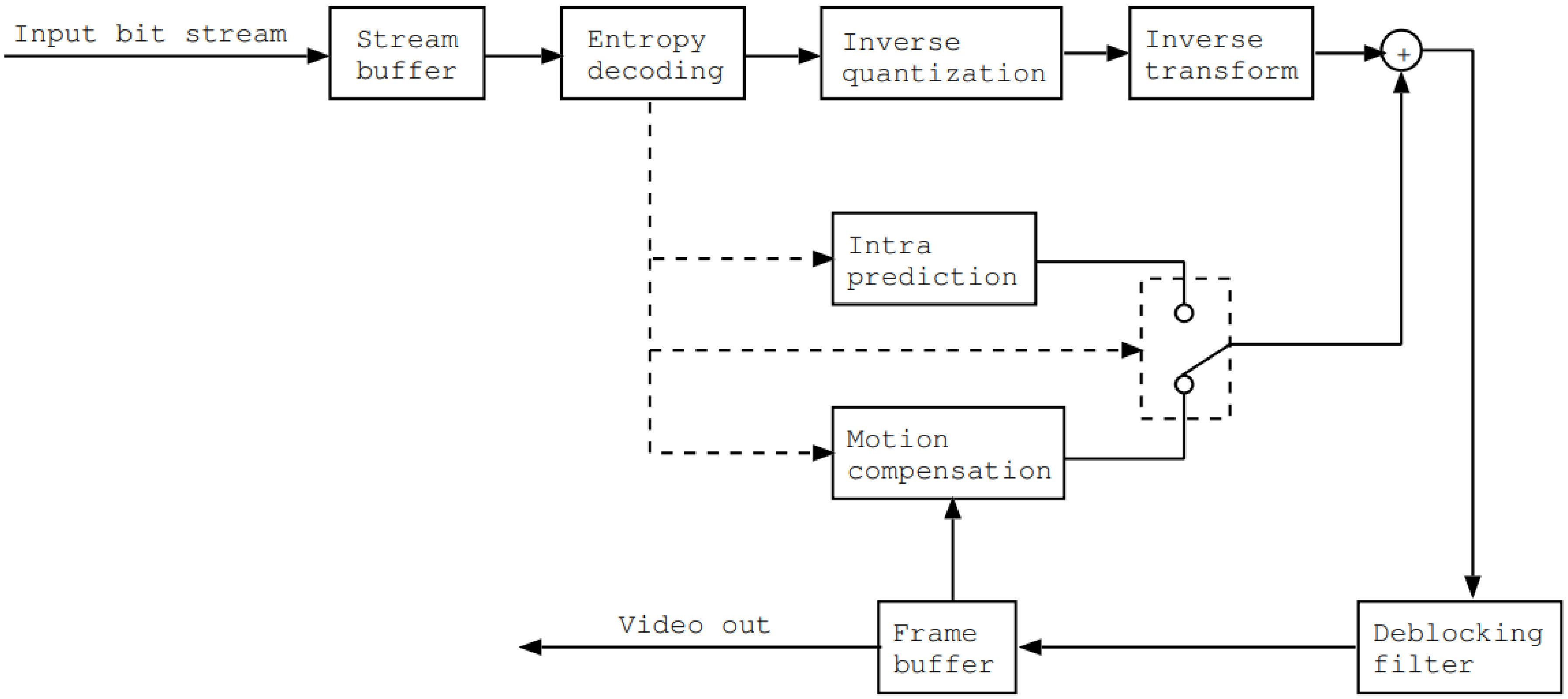
Theoretical background

AV1 compared to H.264

It is similar to H.264 but more difficult and complex.

H.264	AV1
Easy image postprocessing, only using Deblocking Filter	More complicated filtering using Constrained directional Enhancement Filter and Deblocking Filter
Simple image interpolation with 6-tap filter and bilinear interpolation with hard-coded coefficients	8-tap interpolation filter with 16 different coefficients in each of 6 sets of coefficients with a total of **96 different combinations**
Only two methods of inverse transform: IDCT and WHT	DCT-II, DCT-III, ADST-IV, FlipADST, IDHT, WHT
Not optimized for multicore execution	Optimized for multicore execution
Maximum 2 reference frames	Up to 9 reference frames with hierarchy

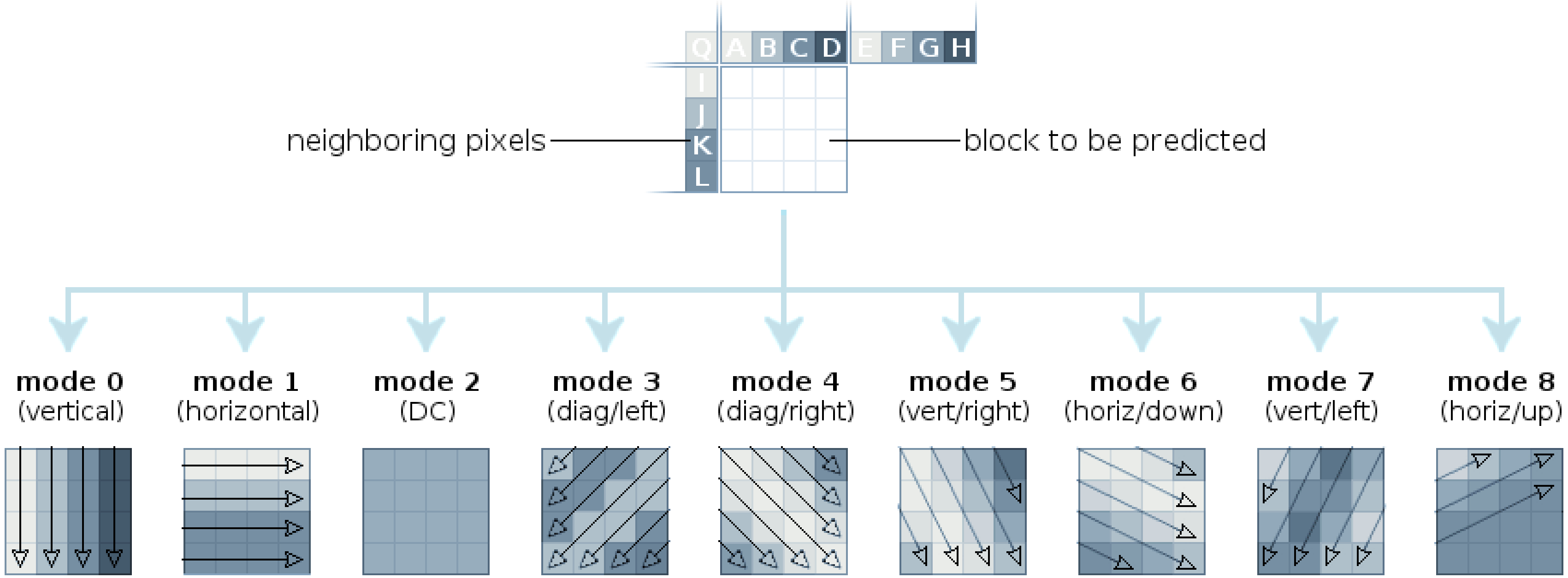
Simplified structure of a video decoder



Intra prediction

Intra prediction is a pixel-by-pixel prediction.

Realized through Intra-coding method, consisting of 56 possible directions.



AVC/H.264 intra prediction modes

Inter prediction (motion compensation)

Inter prediction is a frame-by-frame prediction and is one of the most significant methods of the video size reduction.

Realized through encoding pixel movement with Motion Vectors.

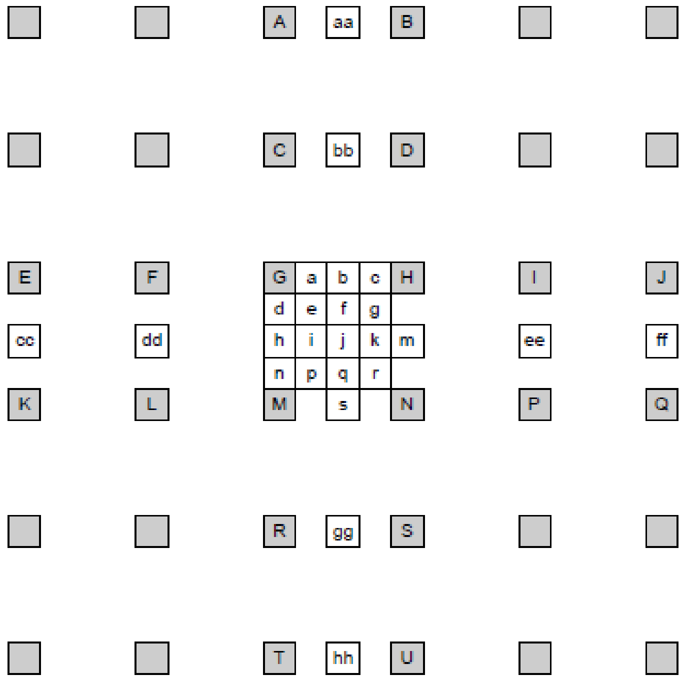
Motion Vectors use reference frames for estimating pixel offset in X and Y dimensions.

Methodological base

Interpolation

Key point: it is very expensive.

8 coefficients 1d interpolation is the product of two vectors consisting of 8 numbers which is about **48 CPU cycles** provided that multiplication takes 4 cycles and addition takes one cycle.

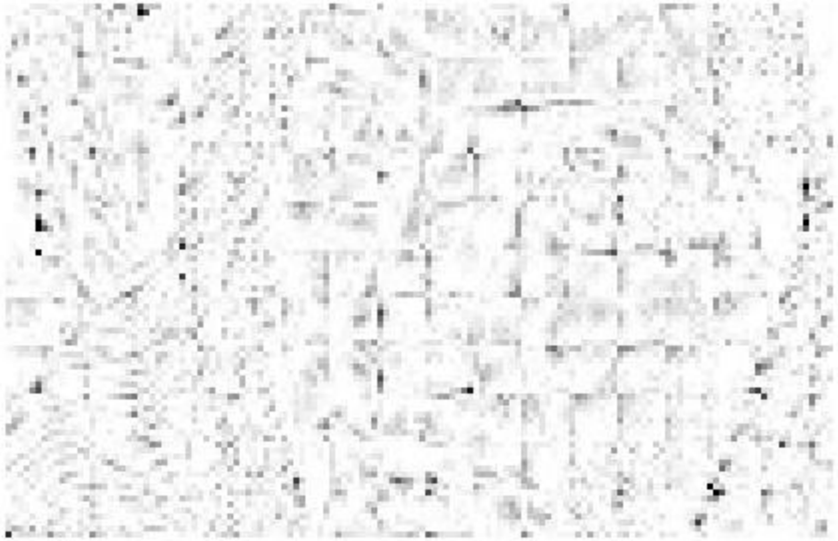


DF (Deblocking Filter)

Key point: it is unnecessary in ultralow resolution.

DF is a method of reducing the blockiness of an image but is unnecessary in 64px resolution.

Closeup of reconstructed image

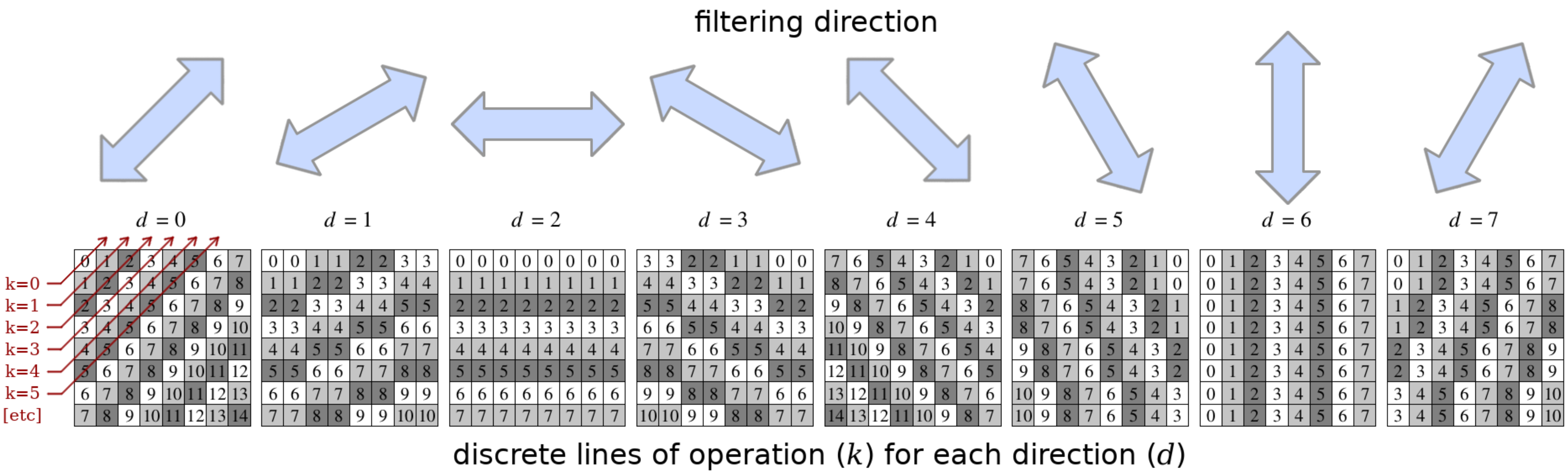


Normalized error distribution within each block

CDEF (Constrained Directional Enhancement Filter)

Key point: it is too expensive operation.

CDEF is applied after the DF and is being used to make transitions between the canvas smoother.



Inverse transformations

Key point: it may be optimized for low resolution.

Inverse transformations such as IDCT are highly optimized but we may optimize it even more, excluding high frequencies.